

100 Days of Azure DevOps

Day 1 Handout — Architecture + Step-by-step Lab

Learning in public · Personal lab only · Educational content · Not a sales pitch

What you will understand today

- **IaaS vs PaaS vs SaaS** — who is responsible when something breaks
- **Region vs Availability Zone** — city vs building
- **Personal Azure setup** — subscription, budget alert, first resource group

High-level architecture

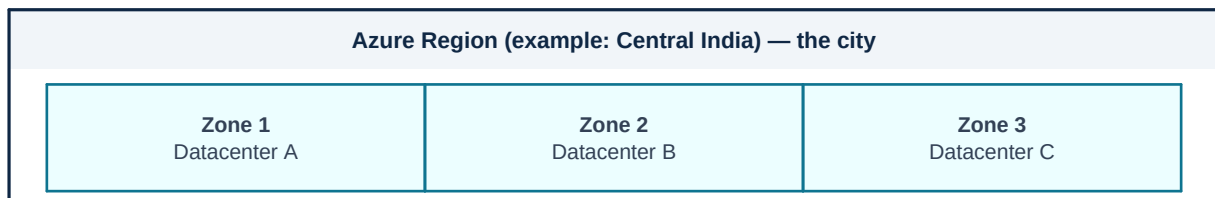
Architecture A — Shared responsibility (who holds the spatula?)

Layer	IaaS (VM)	PaaS (App Service)	SaaS (M365/GitHub)
Application / data	You	You	Vendor
Runtime / middleware	You	Vendor	Vendor
OS / patching	You	Vendor	Vendor
Virtualization / hosts	Vendor	Vendor	Vendor
Physical datacenter	Vendor	Vendor	Vendor
Azure examples	Virtual Machines	App Service, Functions, Azure SQL	Microsoft 365, GitHub

Architecture B — Azure geography map

Concept	What it is	Restaurant metaphor	Why it matters
Geography	Area with compliance boundary	Country / market	Data residency rules
Region	Set of datacenters in a place	City	Latency, DR pairing
Availability Zone	Separate datacenter in a region	Different building, same city	Survive building failure
Resource Group	Logical folder for resources	One kitchen project folder	Lifecycle + cost scope

Architecture C — Region contains Availability Zones



Put critical apps across zones in the same region for higher availability. Use a paired region for disaster recovery across cities.

One-liner to remember

Region = which city · Zone = which building · IaaS/PaaS/SaaS = who holds the spatula

Step-by-step lab (20–30 min)

Use a **personal** Microsoft account and personal Azure subscription. Do not use employer SSO, work tenants, or work data.

- 1 Sign in at **portal.azure.com** with your personal account.
- 2 Open **Subscriptions** → note your subscription **name** and **ID**.
- 3 Open **Cost Management + Billing** → Budgets → create a budget alert (example: Rs 500 or \$20) with email to yourself.
- 4 Create a Resource Group named **rg-day01-lab** in **Central India** (or your nearest region).
- 5 Optional: open **Cloud Shell** and run the commands on the next section.
- 6 Browse a free-tier create page (Storage / App Service) but **do not deploy** anything costly yet.
- 7 Optional cleanup: delete **rg-day01-lab** if you created nothing inside it, or keep it for Day 2.

Commands (optional)

```
az account show --output table
az group create --name rg-day01-lab --location centralindia
az group list --output table
```

Done checklist

- I can explain IaaS / PaaS / SaaS with one example each
- I can explain Region vs Availability Zone
- Budget alert exists on my personal subscription
- Resource group rg-day01-lab exists (or I know how to create it)

Tomorrow — Day 2

Azure Portal vs CLI vs PowerShell — when to click, when to script.